



CortexCall

EEG Motor Imagery Classification — Kickoff

Reading LEFT vs. RIGHT hand imagination straight from brain signals — no movement required.

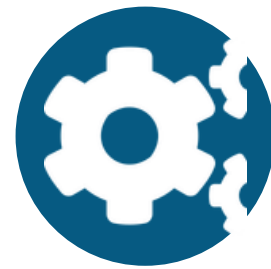
The Big Picture

What we're building and how the whole pipeline fits together

Goal: Classify whether someone is imagining LEFT or RIGHT hand movement — using only their EEG signal, ~2000 recorded trials.



Data



Process



EDA



Model



Deploy

We are here — EDA decides what the rest of the pipeline is built on.

Typical channels: FZ, C3, CZ, C4 (more possible) **Target users:** research team + future assistive-tech demo

Understanding the Dataset

~2000 Excel files. Here's what's actually inside one of them.

time ↓ channels →

	FZ	C3	CZ	C4
t1	-0.93	-0.04	-0.13	0.51
t2	0.34	0.29	-0.99	0.69
t3	0.40	-0.12	0.36	0.24
t4	-0.45	0.86	-0.06	-0.18

label for whole file: LEFT or RIGHT

- **Columns = channels**

Each column is one electrode (FZ, C3, CZ, C4...)

- **Rows = time samples**

Recorded at 128–256 Hz, over 3–8 seconds

- **One label per file**

The whole trial is LEFT or RIGHT — not per-row



Watch this: we must keep a clean filename → label mapping (CSV or in the filename itself) across all 2000 files. If this breaks, everything after it is compromised.



The Signal We're Hunting

Everything else in this project only matters if this shows up.

8–12 Hz

Mu rhythm

13–30 Hz

Beta rhythm

C3 / C4

Motor cortex channels

ERD (Event-Related Desynchronization): power drops in these bands over C3/C4 when someone imagines moving — that dip is the “tell” we're looking for.

Why it's the gate, not just a chart: if PSD plots show no LEFT vs. RIGHT difference here, no model — classic or deep learning — will fix that. We check this before investing in modeling.

This Week: EDA Only

All four of us, working in parallel, on the same raw data.

Week 1

Focus



No sequencing needed within EDA

All 4 people can start immediately — everyone reads the same raw files.



Different question each

We split by question, not by person, so nobody duplicates work.



One shared conclusion

Findings converge into a single summary that Processing will depend on next.

By Friday: we should know whether the data is usable, and have the exact filter band, rejection rule, and epoch window Processing will use next week.

Split by Question, Not by Person

Four notebooks, four different questions — not four copies of the same analysis.



Person A

Class balance, sampling-rate consistency, missing values & NaNs



Person B

Time-domain plots — LEFT vs. RIGHT, raw vs. filtered, on C3/C4



Person C

PSD + ERD analysis on C3/C4 — the “does this data even work” question



Person D

Artifact & outlier statistics — how many trials would we reject, and why



The Risk We Need to Avoid

If everyone cleans the raw data differently, our conclusions stop being comparable.

- Person A drops NaN rows one way
- Person B resamples the signal differently
- Person C uses different channel names

Result: everyone's “conclusions” are actually about different data — comparing apples answering different questions on different apples. We can't converge on one config from that.

The fix is on the next slide — one shared loader, used by everyone.

The Fix: One Shared Loader

The 80/20 rule: most work stays in notebooks — the reusable 20% graduates to a script.



80% — Notebooks

- Plotting, PSD, exploration
- “Let me try X and see” experiments
- Personal, one-off, fast iteration
- Stays individual — not imported by others



20% — `data_loader.py`

- Standardizes channel names/order
- Attaches LEFT/RIGHT label
- Verifies sampling rate — flags mismatches
- No filtering yet — that's next week's decision
- Everyone imports it, nobody copy-pastes it

Shared vs. Individual — At a Glance

Frozen & Shared	Individual per Notebook
File loading logic	Specific plots & analysis
Column / channel naming	Exploratory filtering (just to look, not applied)
Label attachment (filename → LEFT/RIGHT)	Outlier / rejection proposals (findings, not decisions)
Sampling-rate & length verification	Interpretation and conclusions

Rule of thumb: will more than one person import this exact logic? If yes → it belongs on the left.

Project Structure

Folders mirror the pipeline stages, not the people.

cortex-call/

data/

(gitignored — raw + processed, local only)

src/

data_loader.py, config.py — shared, imported code

notebooks/

eda_*.ipynb + eda_summary.md — individual + converged

models/

(gitignored — trained weights)

app/

Streamlit demo (deployment stage)

reports/

final report, slides, figures



src/config.py

The one file every stage depends on — worth guarding with review.



notebooks/eda_summary.md

Converged conclusions — feeds config.py next phase.



data/ & models/

Gitignored, but the folders exist with a note on where to download from.

Team Workflow Rules

Lightweight, not enterprise — just enough to catch the mistakes that matter here.



No direct pushes to main

main stays stable at all times



One branch per task

e.g. eda-psd, eda-class-balance, csp-svm-baseline



PR + 1 review before merge

Team lead reviews, especially anything touching config.py



Flag preprocessing changes in the PR

“Changed epoch window from X to Y” — the thing most likely to silently break comparability



Why Subject-Aware Splits Matter

A detail worth getting right before modeling starts.

✗ Random split

Subject 7's trials land in both train AND test. The model partly learns “what Subject 7's brain looks like,” not LEFT vs. RIGHT. Accuracy looks great — falsely.

✓ Subject-aware split

Train on Subjects 1–16, test only on Subjects 17–20 — people the model has never seen. An honest estimate of real-world performance.

Action item for EDA: check whether our ~2000 files carry a subject ID anywhere — filename, metadata sheet, or folder structure. This decides which CV strategy we can actually use.

Repo Setup

What we're calling it, and what stays out of git.



CortexCall

Public repo · no license yet

EEG-based motor imagery classification (LEFT vs. RIGHT hand) using signal processing, CSP+SVM baseline, and EEGNet deep learning, deployed as a Streamlit demo.

```
cortex-call (lowercase-hyphenated is the safe
convention across OSes)
```

.gitignore essentials

```
# Data
data/
*.xlsx
*.csv

# Models
*.pt
*.pkl
*.joblib
*.onnx

# Python
__pycache__/
venv/ .venv/
.ipynb_checkpoints/

# OS / Editor
.DS_Store
.vscode/
```




Roadmap

Where we're headed after this week

✓ Repo & structure set up

2 Shared data_loader.py

3 EDA — 4 parallel notebooks

4 Converge into eda_summary.md → config.py

5 Preprocessing pipeline

6 CSP + SVM baseline

7 EEGNet model

8 Streamlit demo